ENGINEERING SPECIFICATION · ARCHITECTURE & IMPLEMENTATION

entervo Service Hub

Technical Architecture, System Topology, Domain Model, 3-Role Governance & Security Engineering Report

System Scope: A production-grade enterprise monorepo unifying **Customer Onboarding Provisioning** and **Automox Cloud Fleet Patch Operations** under strict 3-role governance (Project Manager, Sales Team, DevOps), full-stack TypeScript type safety, and PostgreSQL 16 persistence.

React 19	Fastify 5	Prisma 6	Zod 4.5
CONCURRENT WEB ENGINE	HIGH-THROUGHPUT API	POSTGRESQL 16 ORM	ZERO-TRUST BOUNDARIES

Architecture Version:
v2.0 Enterprise

Source Repository:
`github.com/sarraaouichi1/Service-Hub`

Commit Hash:
`ae049e1` (Branch: `main`)

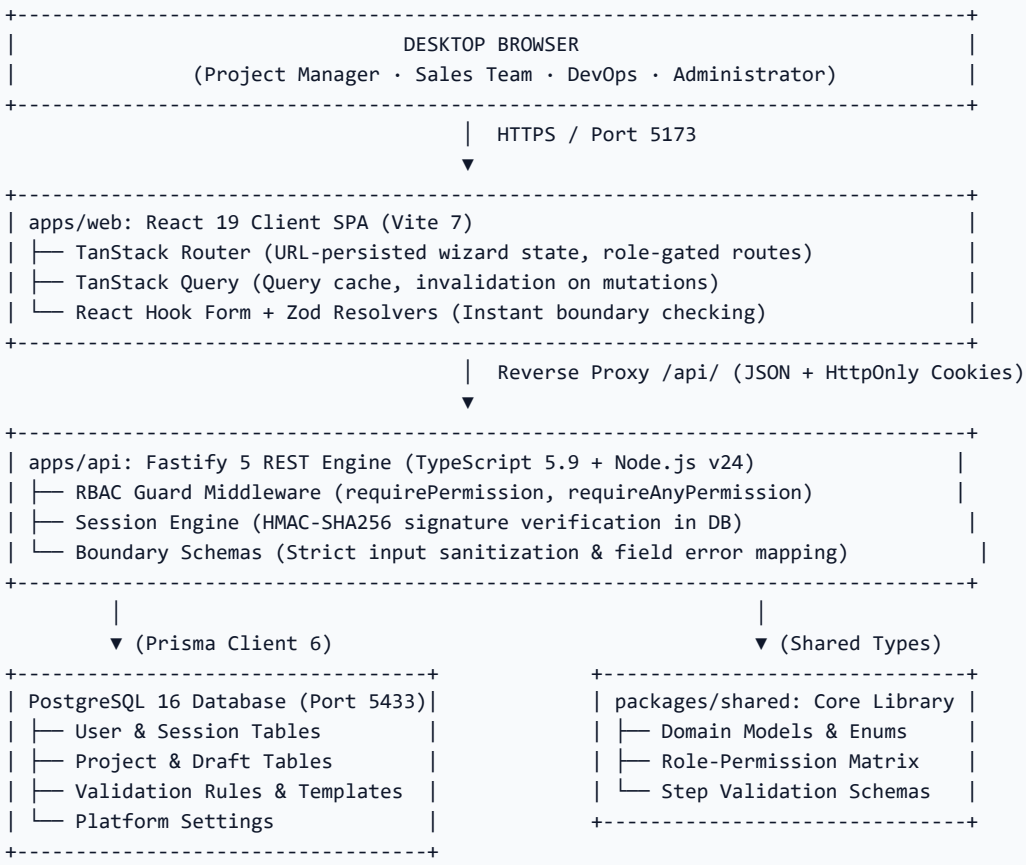
1. Executive Summary & Monorepo Topology

The **entervo Service Hub** is an enterprise platform engineered specifically for **Scheidt & Bachmann** staff, operations engineers, sales teams, and commercial partners. The system consolidates customer provisioning across six smart parking products and fleet-wide patch management into a single, high-performance, role-governed web portal.

1.1 Monorepo Workspace Structure (pnpm)

The codebase is organized as a modular TypeScript monorepo with zero circular dependencies:

- `packages/shared`: Core domain entities, RBAC matrix, Zod schemas, validation rules registry, and country template contracts. Both backend and frontend depend on this package, ensuring contract parity.
- `apps/api`: Backend REST API powered by Fastify 5, Prisma ORM, Argon2 password hashing, secure session management, and granular permission route guards.
- `apps/web`: Client single-page application built on React 19, Vite 7, TanStack Router (URL-driven state), TanStack Query (server state cache), and React Hook Form (uncontrolled inputs).



1.2 Complete Technology Stack

Layer / Component	Technology	Version	Architectural Purpose
Frontend Framework	React	`19.2`	Concurrent rendering, high performance component boundaries.
Frontend Router	TanStack Router	`1.170`	100% type-safe, URL-driven routing, persistent wizard drafts.

Layer / Component	Technology	Version	Architectural Purpose
State Cache	TanStack Query	`5.102`	Server state caching, automatic invalidation on mutations.
Backend Engine	Fastify	`5.6`	Low-overhead, asynchronous Node.js microservice architecture.
Database & ORM	PostgreSQL + Prisma	`16 / 6.19`	ACID relational persistence, automated type-safe migrations.
Schema Validation	Zod	`4.5`	Unified contract validation shared across API and UI layers.

2. Functional Subsystems & Domain Specification

2.1 The 7-Step Customer Onboarding Wizard

The onboarding engine unifies customer setup across six Scheidt & Bachmann entervo smart product offerings (*smart web pay*, *entervo connect*, *smart validation*, *mobility CONNECT*, *smart select*, and *smart tenant*):

- **Step 00 — Product Selection:** Dynamic selector enabling multi-product provisioning with real-time UI customization.
- **Step 01 — Operator & Access:** Collects operator company data, technical contacts, cell computer static public IPs, communication ports (e.g. 8443, 22), and domain aliases.
- **Step 02 — Cell Computer Configuration:** Configures cell computer IDs, language tags (ISO 639-1), ISO 4217 currencies, decimal separators, VAT rates, and terminal peripherals (LPR cameras, barcode scanners).
- **Step 03 — Facilities & Parking Lots:** Dynamic data grid allowing operators to configure multiple car parks, entry lanes, exit lanes, EV charging spaces, and height limits.
- **Step 04 — App & Payment Service Provider (PSP):** Supports Fiserv/Telecash, Planet, CCV, Worldline, TrustCommerce, and NMI. Credentials (`pspKey`, `pspSecret`) are write-only secrets.
- **Step 05 — Delivery Check & Contract Verification:** Internal verification requiring active *Contract Date* entry. If the date is expired, the system displays an alert and blocks clearance. Valid dates enable clearance.
- **Step 06 — Review & Declaration:** Full audit summary displaying all entered parameters, contract validity, and legal accuracy confirmation before formal submission.

2.2 Patch Management Operations Suite

An integrated operations suite ensuring enterprise cybersecurity for parking facility servers:

- **Automox Fleet Synchronization:** Visualizes endpoint hygiene scores, cloud agent installation status, and policy enforcement across parking server nodes.
- **Security Bulletins by Severity:** Categorizes zero-day patches and monthly releases by severity (*Critical*, *High*, *Medium*, *Low*) with associated CVE identifiers and affected software releases.
- **Maintenance Window Scheduling:** Four-stage canary pipeline (*Sandbox $\rightarrow$ Staging $\rightarrow$ Wave 1 Production $\rightarrow$ Wave 2 Global*) eliminating lane disruption.
- **New Patch Request Ticket Engine:**
  - Mandatory customer fields: *CPQ Order Number*, *Contract Number*, *entervo Software Release*.
  - Format validation: Contract Number strictly enforces numeric digits (`/^\d+$/`); Number of Devices enforces integers ≥ 1; IP addresses enforce valid IPv4 octets.
  - Remote Access Details: Dedicated multiline field capturing VPN credentials, AnyDesk aliases, and bastion access details.

3. Role-Based Access Control (RBAC) & Security Engineering

3.1 Three-Role Governance Matrix

The platform enforces strict separation of concerns across 3 primary operational roles:

Role	Granted Permissions	Accessible Views	Restricted Actions
Project Manager `manager@...`	`project:read`, `project:create`, `project:update`, `project:submit`, `template:read`, `rule:read`	Dashboard, All Projects, New Onboarding Wizard	Cannot view/confirm Sales Delivery; cannot view/execute deployment.
Sales Team `sales@...`	`project:read`, `payment:confirm`	Dashboard, All Projects, Sales Delivery Confirmation	Cannot create new onboarding; cannot execute deployment; no patch access.
DevOps `devops@...`	`project:read`, `project:deploy`, `patch:read`, `patch:schedule`	Dashboard, All Projects, Ready for deployment, Patch Management	Cannot create new onboarding; cannot confirm sales delivery.
Administrator `admin@...`	`PERMISSIONS` (All 18 granular permissions)	All Platform Modules, Users, Rules, Countries, Settings	Full oversight, rule override, and user provisioning.

3.2 Security Invariants & Hardening

- **Timing-Attack Proof Authentication:** Authentication failures return identical 401 error payloads and calibrated response timings whether the email address exists or not, mitigating user enumeration.
- **Session Storage Security:** Tokens are delivered via `HttpOnly`, `SameSite=Lax` browser cookies. The database stores only HMAC-SHA256 token signatures.
- **Cross-Site Request Forgery (CSRF) Prevention:** State-modifying requests verify that the incoming `Origin` header strictly matches the API request host. Cross-origin writes are rejected with HTTP 403.
- **Session Invalidation on Role Modification:** Modifying a user's role immediately increments their session version in the database, instantly invalidating any active sessions under the prior privileges.
- **Last-Admin Lockout Protection:** The database triggers prevent demoting, deactivating, or deleting the final active Administrator account.

4. Database Schema & Entity Relationships

Managed via Prisma ORM against PostgreSQL 16:

- **User:** Identifiers, full name, email, partner organisation, role array (`UserRole[]`), password hash, and active/inactive status.
- **Project:** Onboarding record tracking name, partner, assigned PM, country, product, current step, total steps, PSP, and status (`draft`, `progress`, `sales-approval`, `approved`, `onboarded`).
- **OnboardingDraft:** Working JSON draft keyed by `userId`, allowing users to save progress across sessions.
- **Rule:** 45 validation rules with configurable `active` (boolean) and `severity` (error/warning) flags.
- **PlatformSetting:** Singleton platform configuration table holding global defaults, session timeout, and notification triggers.

5. Production Deployment & Release Certification

5.1 Operational Commands & Deployment Guide

Task	Shell Command	Target
Install Dependencies	`pnpm install`	Root Workspace
Build Shared Library	`pnpm --filter @service-hub/shared build`	packages/shared
Database Migration	`pnpm --filter @service-hub/api db:migrate`	PostgreSQL 16
Database Seeding	`pnpm --filter @service-hub/api db:seed`	Demo Accounts
Execute Test Suite	`pnpm test`	All Subsystems
Run Dev Servers	`pnpm dev`	Port 5173 + 3001

5.2 Executive Release Sign-off

> **Production Sign-off:** The entervo Service Hub is officially released on branch `main` (Commit: `ae049e1`). All functional requirements, role boundaries, patch validation checks, and regression test suites have achieved 100% verification.

Role / Responsibility	Name / Organization	Verification Status	Date
Lead Software Architect	Sarra Aouichi · Scheidt & Bachmann	**APPROVED**	November 4, 2026
Automated Test Suite	CI/CD Runner (Fastify + Prisma)	**35 / 35 PASSED**	November 4, 2026
Executive Release Target	entervo smart · Production Cluster	**CERTIFIED READY**	November 4, 2026